

Assignment: Visualization & Data Preparation

Dataset

You will find the **credit_customers** dataset on the drive. Use it to solve this assignment.

1. Exploratory Data Analysis (EDA)

Perform a brief initial exploration of the dataset.

- Display the number of rows, columns, and data types.
- Show descriptive statistics (mean, min, max, std) for numerical features.
- **Visualization Task:**
 - Create a **histogram** of age to understand the age distribution of applicants.
 - Create a **bar chart** of purpose to show the most common loan purposes.

2. Handling Missing Values

- Identify missing values in the dataset.
- Choose and apply an appropriate handling technique (imputation or deletion)
- Explain why you chose this technique..
- **Visualization Task:** Before and after handling missing values, create a **bar chart** that shows the count of missing values per column.

3. Handling Duplicated Data

- Check if there are duplicated rows.
- Remove them if they exist.
- **Visualization Task:** Show a **before vs. after comparison** of dataset size using a simple **bar chart** or **text-based comparison**.

4. Outlier Detection and Handling

- Detect outliers in **numerical columns** (e.g., credit_amount, age, duration) using a statistical method (like IQR or Z-score).
- **Visualization Task (Step 1):** Use **boxplots** to visualize outliers in at least **two numerical features**.
- Handle the outliers (remove or impute).
- **Visualization Task (Step 2):** Show the boxplots again after handling outliers to compare the effect.

5. Data Transformation

- Apply feature scaling (Min-Max scaling or Z-score normalization) to **numerical features**.
- **Visualization Task:** Show the distribution of one numerical feature before and after scaling using a **histogram**.
- Convert categorical variables into numerical format using **one-hot encoding** or **label encoding**.

6. Class Imbalance

- Check the balance between classes in the target variable (class).
- **Visualization Task:** Create a **bar plot** to show the distribution of "good" vs "bad" classes.
- If imbalance exists, apply a strategy to handle it.
- **Visualization Task (after balancing):** Show the updated class distribution with another **bar plot**.

7. Splitting the Data

- Split the data into training and testing sets.
- Try at least **two different split strategies** (e.g., 70/30 vs. 80/20).